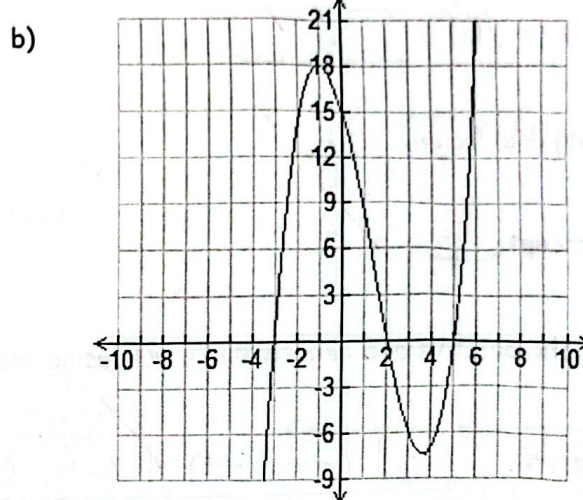
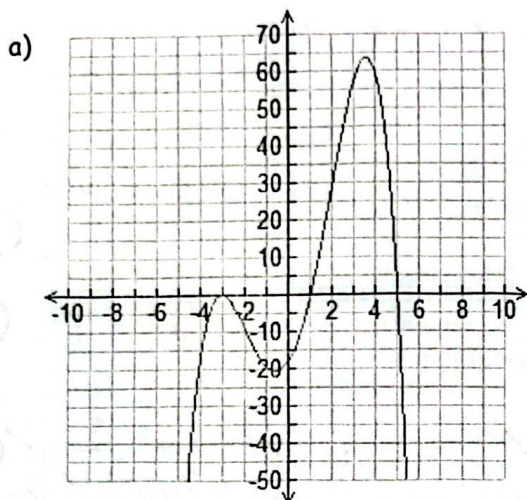


Name _____

Date Sept. 12

KNOWLEDGE

1. Given the following graphs, complete the chart below:



Graph	Odd/Even Degree	Least Possible Degree	End Behaviour	# of turning points	# absolute minimums & maximums	# of local minimums & maximums
a)	even	4	$x \rightarrow -\infty, y \rightarrow -\infty$ $x \rightarrow \infty, y \rightarrow -\infty$	3	1	3
b)	odd	3	$x \rightarrow -\infty, y \rightarrow -\infty$ $x \rightarrow \infty, y \rightarrow \infty$	2	n/a	2

including 1 absolute max

(6)

2. The table of values represent a polynomial function. Determine the:

a) Value of the constant finite differences -24b) Degree of the polynomial function 3c) Sign of the leading coefficient -d) Value of the leading coefficient -4

x	y
-3	124
-2	41
-1	8
0	1
1	-4
2	-31
3	-104
4	-247

-83
 50
 -33
 -7
 -5
 -27
 -73
 -143

-24
 -24
 -24
 -24
 -24
 -24
 -24

$$FD = a \cdot n!$$

$$-24 = a \cdot (3 \times 2 \times 1)$$

$$a = \frac{-24}{(6)}$$

=

$$f(x) = (-x+2)(x+4)^2(x-1)$$

$$= -(\overset{+2}{x-2})(\overset{-4}{x+4})^2(\overset{+1}{x-1})$$

3. Given the function $f(x) = (2-x)(x+4)^2(x-1)$, complete the following:

a) Graph the equation:

b) Describe the graph in terms of end behaviour.

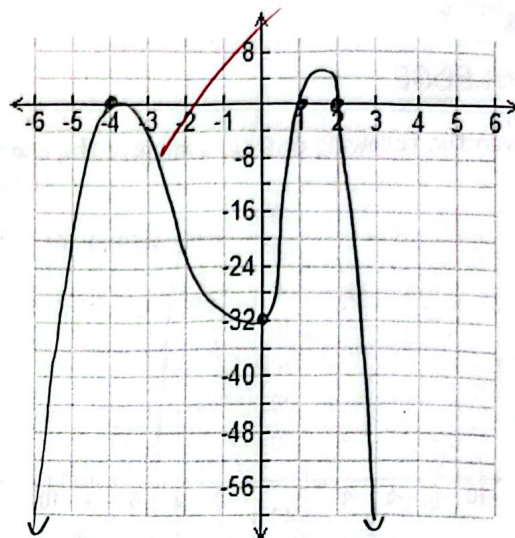
$$x \rightarrow -\infty, f(x) \rightarrow -\infty$$

$$x \rightarrow +\infty, f(x) \rightarrow -\infty$$

c) Leading coefficient: -1

d) y-intercept: -32

e) State the positive and negative intervals using the given chart:



Interval	$(-\infty, -4)$	$(-4, 1)$	$(1, 2)$	$(2, \infty)$
Sign of $f(x)$	-	-	+	-

4. Sketch a polynomial function that satisfies the following conditions:

- $x \rightarrow -\infty, f(x) \rightarrow -\infty$
- even number of x-intercepts, one must be order 3
- three turning points
- 1 absolute maximum

